Day 3 Covariance and Correlation

Complete student notes with worked examples and Python practice

Covariance and correlation describe how two variables move together. These notes connect paired deviations to covariance, standardise covariance into Pearson correlation, introduce rank correlation, and examine scatter plots and the limits of association.

Introduction to paired variables

Variation within one variable describes spread around its own mean. Analysing two variables together reveals whether their values tend to move in the same direction or in opposite directions. Practice hours and assessment scores provide an example of paired observations.

Covariance describes joint variation and its direction. Pearson correlation expresses linear association on a common scale. A scatter plot reveals structure that a single numerical coefficient may miss.

The main dataset

Observation	Practice hours	Assessment score
1	2	35
2	4	48
3	5	55
4	7	68
5	8	74
6	10	90

Each row is one paired observation. The hours and score in a row must refer to the same case. The six illustrative pairs demonstrate the calculation method and do not support claims about a wider student population.

1 Covariance from paired deviations

From variance to covariance

Variance multiplies a variable’s deviation by itself. Covariance multiplies the deviation in X by the paired deviation in Y. This reveals whether the variables tend to be above or below their own means together.

X deviation	Y deviation	Product	Contribution
Positive	Positive	Positive	Both above their means
Negative	Negative	Positive	Both below their means
Positive	Negative	Negative	Opposite sides of their means
Negative	Positive	Negative	Opposite sides of their means

Formulas and variable meanings

Population covariance = sum of (xi - population mean of X) × (yi - population mean of Y), divided by N. Sample covariance = sum of (xi - sample mean of X) × (yi - sample mean of Y), divided by n - 1.

$$\mathrm{Cov}_{pop}(X, Y) = \frac{\sum_{i=1}^{N} (x_i - \mu_X)(y_i - \mu_Y)}{N}$$

$$\mathrm{Cov}_{sample}(X, Y) = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

xi and yi are the paired values in row i. N is the population size; n is the sample size. The same complete pairs must be used for both means and all deviation products. Here we demonstrate both formulas and label each result.

Manual dry run using the main dataset

Mean hours = 36 / 6 = 6. Mean score = 370 / 6 = 61.6667. Values below are rounded for display; keep full precision in the calculation.

Hours x	Score y	x - mean x	y - mean y	Product
2	35	-4	-26.6667	106.6667
4	48	-2	-13.6667	27.3333
5	55	-1	-6.6667	6.6667
7	68	1	6.3333	6.3333
8	74	2	12.3333	24.6667
10	90	4	28.3333	113.3333

Sum of deviation products = 285. Population covariance = 285 / 6 = 47.5. Sample covariance = 285 / 5 = 57. Both are positive, indicating that higher hours tend to occur with higher scores in this dataset.

Covariance units are hours × marks. Its numerical size depends on the units and variability, so 57 is not a percentage or a standardised strength score. Cov(X, X) equals Var(X).

2 Pearson correlation and its interpretation

Why standardise covariance

A covariance changes if hours are expressed in minutes. Correlation divides covariance by the standard deviations of both variables so that the units cancel and the value lies between -1 and +1, provided both variables have nonzero variation.

Sample Pearson correlation: r = sample covariance(X, Y) / (sample standard deviation of X × sample standard deviation of Y). Population correlation uses population covariance and population standard deviations consistently.

$$r = \frac{\text{Cov}_{sample}(X, Y)}{s_X s_Y}$$

Complete the manual calculation

For the six observations, sample variance of hours = 8.4, giving sample standard deviation 2.8983. Sample variance of scores = 387.4667, giving sample standard deviation 19.6842.

r = 57 / (2.898275349 × 19.684172999) = 0.9991208898, approximately 0.9991. This describes a very strong positive linear association in these six observations.

Correlation	Meaning
r = +1	All points lie on an increasing straight line.
r = -1	All points lie on a decreasing straight line.
r near 0	Little linear association in the observed data; inspect the plot.
Larger absolute r	Closer linear association, with context and sample size considered.

The sign gives direction. The absolute value describes linear strength. For example, -0.9 represents a stronger linear association than +0.4. Strength thresholds vary by field and should not be treated as universal rules.

Covariance compared with correlation

Property	Covariance	Pearson correlation
Units	Product of the two units	Unitless
Range	No fixed universal bounds	-1 to +1 when defined
Positive unit rescaling	Changes magnitude	Unchanged
Purpose	Describe joint variation	Describe linear direction and strength

A negative rescaling reverses the correlation sign. Adding a constant to either variable leaves Pearson correlation unchanged. Constant columns have zero standard deviation, so correlation is undefined rather than zero.

3 Ranks and careful analytical conclusions

Pearson and Spearman

Pearson measures linear association between numerical variables. Spearman is Pearson correlation of the ranks and measures monotonic association: whether values consistently tend to increase together or move in opposite directions, even when the relationship is curved.

For x = [1, 2, 3, 4, 5] and y = [1, 4, 9, 16, 25], both rank lists are [1, 2, 3, 4, 5]. Spearman correlation is 1, while Pearson correlation is approximately 0.9811. The relationship increases consistently but is not a straight line.

Ordinal data can be analysed through ranks when its order is meaningful. With ties, pandas rank() assigns average ranks by default. Spearman can reduce the influence of extreme magnitudes, but it is not immune to unusual observations or sampling problems.

Data quality and interpretation

- A scatter plot can reveal a straight line, curve, cluster or unusual observation that affects interpretation.
- The two variables must remain paired. Sorting or filtering complete rows together preserves matching records; sorting each column independently can destroy the association.
- Unusual observations can strengthen, weaken or reverse a correlation. Their treatment requires investigation and a documented reason.
- A descriptive coefficient measures association. It does not show that one variable caused the other.

Zero correlation can hide a clear pattern

For x = [-2, -1, 0, 1, 2] and y = [4, 1, 0, 1, 4], y equals x squared. Pearson correlation is zero because the symmetric U-shaped pattern cancels the linear association. The variables still have an exact mathematical relationship. Zero correlation therefore does not establish independence.

Association and causation

Students with more practice hours might also differ in attendance, prior knowledge or motivation. These factors may contribute to the observed association. A causal claim needs an appropriate design and assumptions; a large correlation alone does not establish it.

Example interpretation

Across six illustrative observations, practice hours and scores have Pearson r = 0.9991 and sample covariance 57 hours × marks. The scatter plot shows a strong increasing linear pattern. This small illustrative dataset does not establish a causal effect or justify generalising to all students.

Extension for later study: r squared equals R squared for ordinary least squares regression with one predictor and an intercept on the same observations. This identity is not a general rule for every predictive model. Inference and p-values will be covered separately.

4 Python covariance and correlation

Run practical examples in order in one notebook or Python session. The same NumPy, pandas and Matplotlib environment used on Day 2 is sufficient.

Step 1 Create and check paired observations

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

hours = np.array([2, 4, 5, 7, 8, 10], dtype=float)
scores = np.array([35, 48, 55, 68, 74, 90], dtype=float)
data = pd.DataFrame({"hours": hours, "score": scores})
print("Shape:", data.shape)
print("Missing values:", data.isna().sum().to_dict())
```

Expected: shape (6, 2), with zero missing values in both columns. data preserves the row pairing. Equal lengths alone do not prove correct matching; verify the source identifiers in a real dataset.

Step 2 Calculate covariance from deviations

```
n = len(hours)
dx = hours - np.mean(hours)
dy = scores - np.mean(scores)
products = dx * dy
cross_total = np.sum(products)
pop_cov = cross_total / n
sample_cov = cross_total / (n - 1)
print("Product total:", cross_total)
print("Population covariance:", pop_cov)
print("Sample covariance:", sample_cov)
```

Expected: 285.0, 47.5 and 57.0, allowing for floating point precision. dx and dy hold deviations; products stores their paired products; cross_total is their sum. n is the number of complete pairs.

Step 3 Verify and standardise

```
cov_matrix = np.cov(hours, scores, ddof=1)
r_manual = sample_cov / (np.std(hours, ddof=1) *
                         np.std(scores, ddof=1))
r = np.corrcoef(hours, scores)[0, 1]
print("Covariance matrix:")
print(np.round(cov_matrix, 4))
print("Manual r:", round(r_manual, 4))
print("NumPy r:", round(r, 4))
```

Expected covariance matrix: [[8.4, 57.0], [57.0, 387.4667]]. Both correlation results round to 0.9991. The covariance matrix diagonal holds variances; [0, 1] and [1, 0] hold the pair covariance. For np.cov(data.to_numpy(), rowvar=False), columns are treated as variables.

5 Python unit changes ranks and edge cases

Step 4 Change hours to minutes

```
minutes = hours * 60
cov_minutes = np.cov(minutes, scores, ddof=1)[0, 1]
r_minutes = np.corrcoef(minutes, scores)[0, 1]
print("Covariance with minutes:", cov_minutes)
print("Correlation with minutes:", round(r_minutes, 4))
print("Pandas correlation:")
print(data.corr().round(4))
```

Expected covariance 3420.0 minute × marks, or 60 × 57. Correlation remains 0.9991. The pandas correlation matrix has 1 on its diagonal and 0.9991 off the diagonal. Positive unit changes do not alter the linear association.

Step 5 Calculate Spearman through ranks

```
x_rank_demo = pd.Series([1, 2, 3, 4, 5])
y_rank_demo = pd.Series([1, 4, 9, 16, 25])
pearson_demo = x_rank_demo.corr(y_rank_demo)
x_ranks = x_rank_demo.rank()
y_ranks = y_rank_demo.rank()
spearman_demo = x_ranks.corr(y_ranks)
print("Pearson:", round(pearson_demo, 4))
print("X ranks:", x_ranks.tolist())
print("Y ranks:", y_ranks.tolist())
print("Spearman:", round(spearman_demo, 4))
```

Expected Pearson 0.9811; both ranks [1.0, 2.0, 3.0, 4.0, 5.0]; Spearman 1.0. x_ranks and y_ranks replace magnitudes with ordered positions. The rank correlation uses pandas and does not require a separate statistics package.

Step 6 Handle a constant column explicitly

```
constant = np.array([60, 60, 60, 60, 60, 60])
if np.std(constant) == 0 or np.std(hours) == 0:
    print("Correlation is undefined: a variable is constant.")
else:
    print(np.corrcoef(hours, constant)[0, 1])
```

Expected: the undefined-correlation message. This guard suits the exact example values. A production workflow should also check finite values and whether variation is meaningful at the measurement precision.

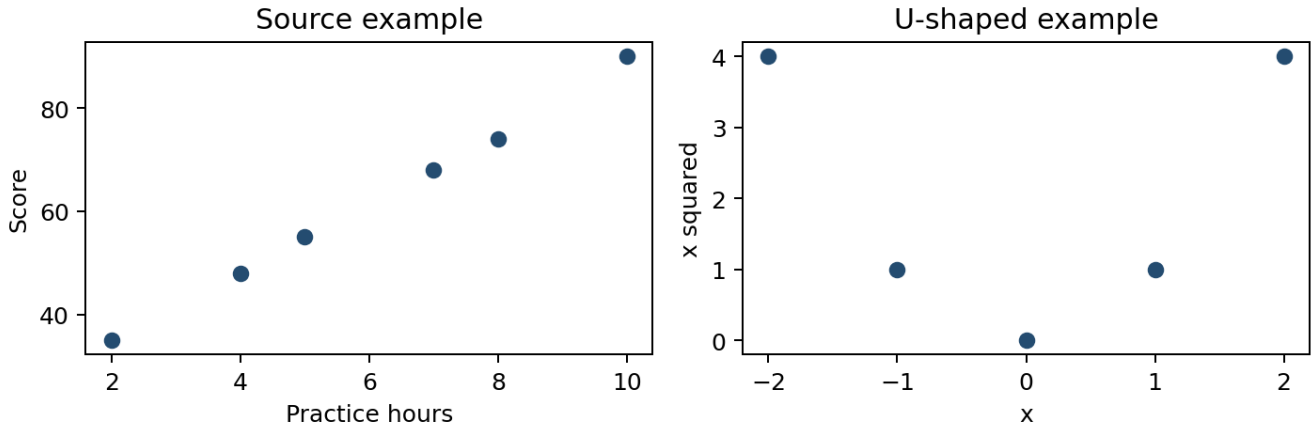
6 Python scatter plots and nonlinear patterns

Step 7 Compare a line-like pattern with a curve

```
curve_x = np.array([-2, -1, 0, 1, 2], dtype=float)
curve_y = curve_x ** 2
curve_r = np.corrcoef(curve_x, curve_y)[0, 1]
print("Curve Pearson r:", round(curve_r, 4))

fig, axes = plt.subplots(1, 2, figsize=(8, 3))
axes[0].scatter(hours, scores)
axes[0].set(xlabel="Practice hours", ylabel="Score",
            title="Source example")
axes[1].scatter(curve_x, curve_y)
axes[1].set(xlabel="x", ylabel="x squared",
            title="U-shaped example")
fig.tight_layout()
plt.show()
```

Expected curve correlation: 0.0. The left plot shows an increasing, nearly straight-line pattern. The right plot has a U shape despite zero Pearson correlation.



Interpreting the figure

Pearson summarises the increasing linear pattern well. On the left half of the U-shaped curve, y falls as x rises; on the right half, y rises. A single linear coefficient misses that structure.

Analysis workflow

Define the two variables and units. Verify matching records and complete pairs. Inspect the scatter plot. Calculate an appropriate coefficient. Investigate unusual values and relevant subgroups. Report the direction, magnitude, sample size and limitations together.

7 Practice exercises and solutions

Exercises

Task 1: For x = [1, 2, 3] and y = [2, 4, 6], find both covariance formulas and Pearson r.

Task 2: Replace y with [6, 4, 2] and explain the sign change.

Task 3: Express hours in minutes and explain why covariance changes.

Task 4: Explain the U-shaped example.

Task 5: Explain what happens if every score is 60.

Solutions

Task 1: means 2 and 4; deviation products [2, 0, 2]; sum 4. Population covariance 4/3, sample covariance 2, r = 1.

Task 2: products [-2, 0, -2]; population covariance -4/3, sample covariance -2, r = -1.

Task 3: covariance multiplies by 60, with the same correlation.

Task 4: a nonlinear symmetric pattern can have zero linear correlation.

Task 5: zero score standard deviation makes Pearson correlation undefined.

Review questions

What do the off-diagonal entries of a covariance matrix mean? Why is correlation unitless? Is -0.9 weaker than +0.4? Can Pearson r be calculated without normally distributed data? Yes: calculating the descriptive coefficient does not require normality; inferential procedures have additional assumptions.